

Study Objective

This study was inspired by stakeholder discussions as part of the Harvest Innovation Grant, where Eagle County School District (ECSD) is exploring opportunities for a farm-to-school program. During these meetings, this parcel (#193926100007; Red Mountain Ranch, ~40 acres) was identified as a potential site for agricultural use, despite not being actively farmed for approximately 10 years.

In a recent meeting, local ranchers noted that the parcel was once much greener, but now appears more brown, suggesting a decline in vegetation condition over time. This study uses remote sensing to evaluate the observed change and provide spatial evidence to support these discussions and inform potential agricultural restoration.

Specifically, this study compares summer vegetation conditions across three time periods (2005, 2015, and 2025) using satellite-derived NDVI to assess whether vegetation health has declined or improved over time and to identify potential for recovery.

Methods

The study area was defined using Eagle County parcel data obtained from the Eagle County ArcGIS Online. The parcel of interest was isolated in ArcGIS Pro using a “Select by Attribute” query (PARCEL_NUM = 193926100007) and exported as a shapefile for analysis.

Satellite imagery was accessed and processed using Google Earth Engine (GEE). Landsat imagery was selected to represent average summer vegetation conditions using a composite of images between June 1st and August 31st for each year. The following datasets were used:

- Landsat 5 Surface Reflectance (Summer 2005)
- Landsat 8 Surface Reflectance (Summer 2015)
- Landsat 8 Surface Reflectance (Summer 2025)

Clouds, shadows, and water were removed using the QA_PIXEL band. Surface reflectance values were scaled prior to analysis. Mean composite images were generated for each summer period and clipped to the parcel boundary.

Normalized Difference Vegetation Index (NDVI) was calculated for each composite using:

- Landsat 5: $NDVI = (Band\ 4 - Band\ 3) / (Band\ 4 + Band\ 3)$
- Landsat 8: $NDVI = (Band\ 5 - Band\ 4) / (Band\ 5 + Band\ 4)$

NDVI was used as a proxy for vegetation health, where higher values indicate greener and more productive vegetation.

Mean NDVI values were calculated in Google Earth Engine using a reduceRegion function limited to the parcel geometry. NDVI rasters were exported to ArcGIS Pro for visualization and map layout. Zonal statistics were also calculated in ArcGIS Pro to summarize NDVI values, though GEE values were used as the primary analytical results.

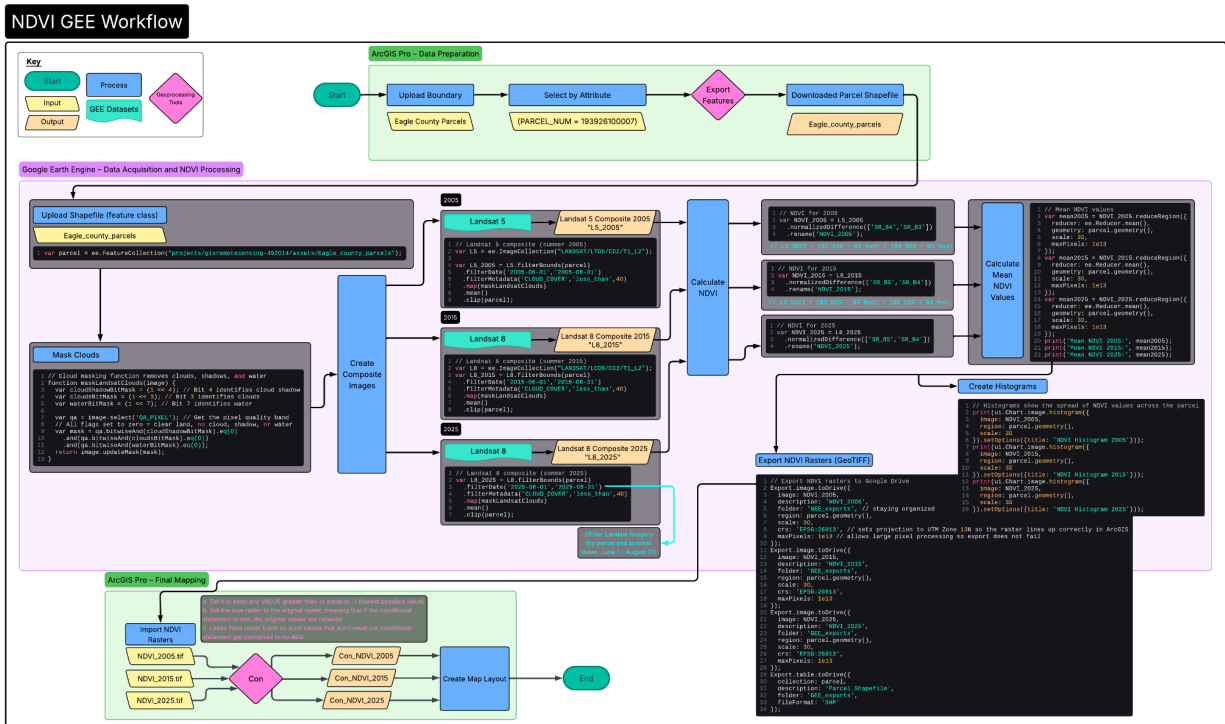
Workflow

To clearly show the analytical process used in this study, two workflow diagrams were created using Lucidchart (lucid.app). One feature I found especially useful is the ability to include code blocks directly in the diagram, which made it easier to organize and understand the workflow.

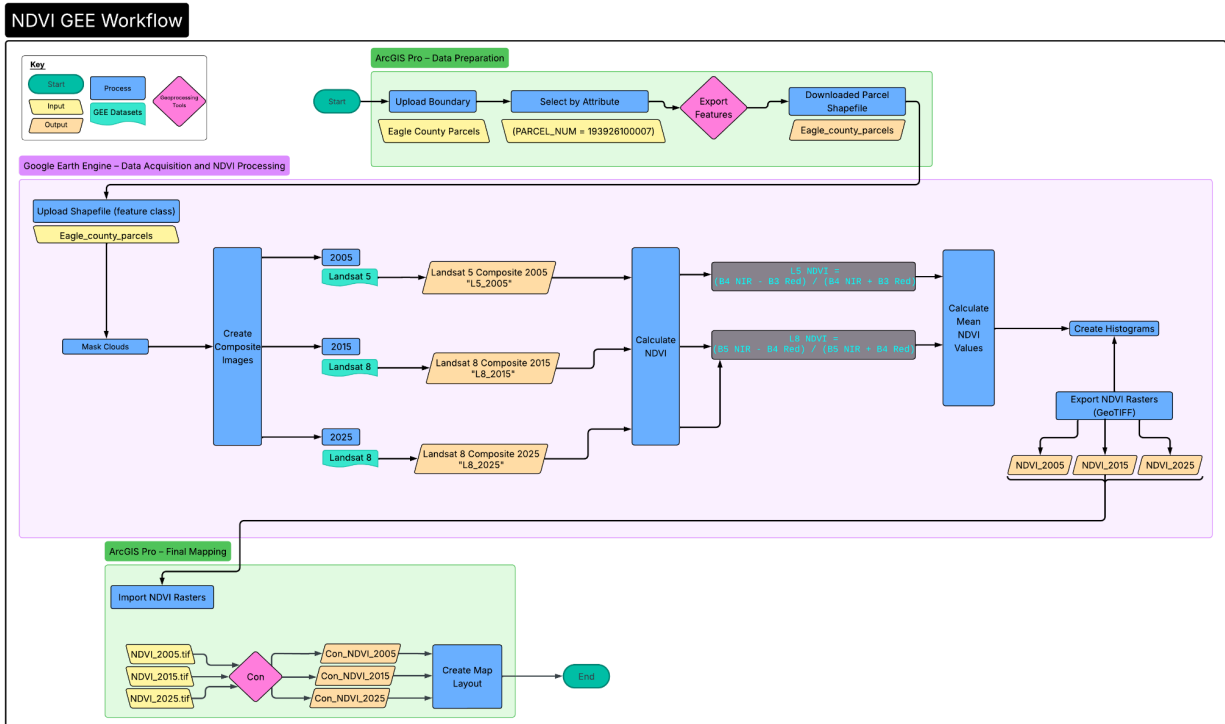
[Figure 1] presents a more detailed version of the workflow that includes key Google Earth Engine code blocks. Organizing the code in this way helped break the script into manageable sections and made it easier to understand how each step contributes to the final outputs.

[Figure 2] provides a simplified version of the workflow, focusing on the main processing steps without code. This version highlights the overall structure of the analysis, beginning with parcel preparation in ArcGIS Pro, followed by image processing and NDVI calculation in Google Earth Engine, and ending with map production.

Together, these diagrams provide both a technical and conceptual overview of the workflow used in this study. They helped me organize my thinking and support replication of the analysis



[Figure 1] - detailed workflow



[Figure 2] - simplified workflow

Results

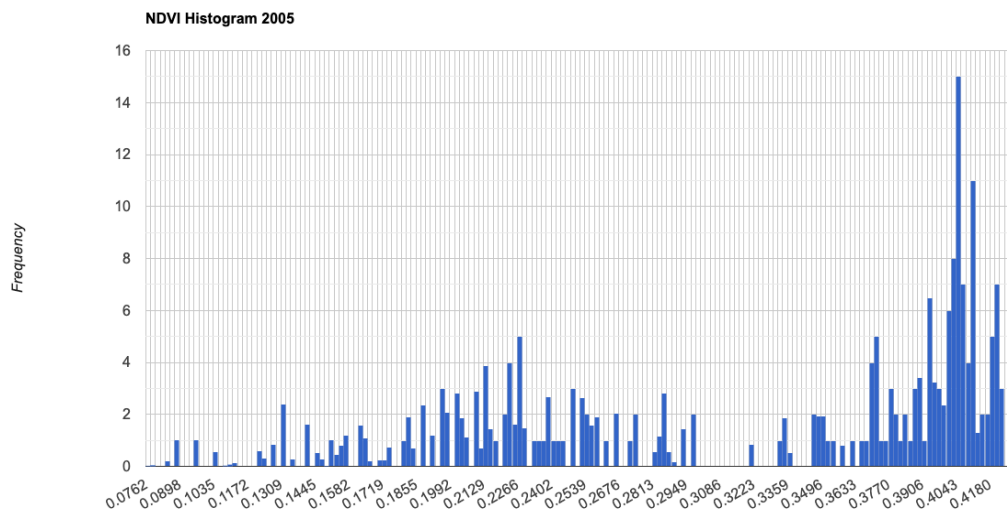
NDVI values across the parcel ranged from approximately 0.08 to approximately 0.30 across all three years, indicating generally sparse to moderate vegetation. No negative NDVI values were observed, suggesting the absence of water or extremely non-vegetated surfaces within the parcel.

Mean NDVI values showed a decline over time:

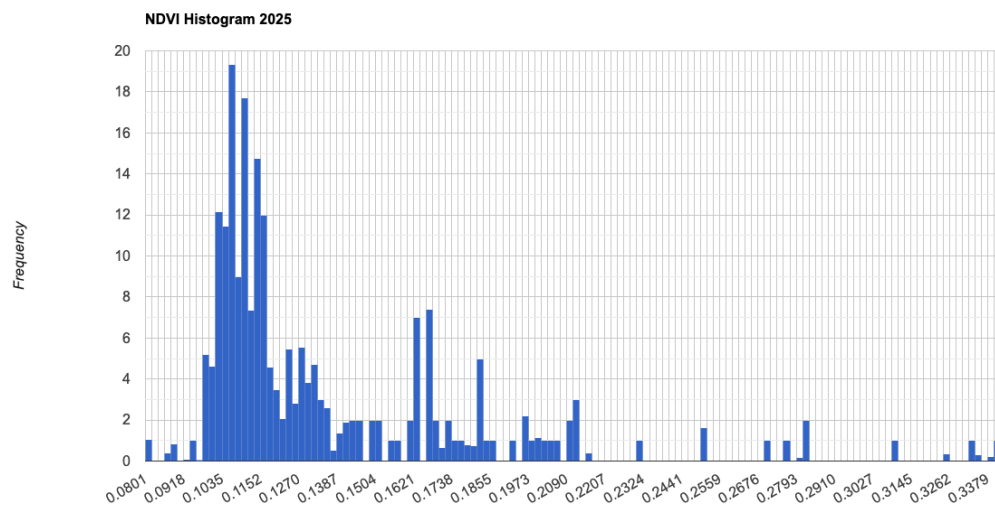
- 2005 = 0.3196529857259515
- 2015 = 0.2261265107619816
- 2025 = 0.13675767482896273

Spatial patterns in the maps indicate that vegetation was more uniformly distributed in 2005, with higher NDVI values across the parcel. By 2015 and especially 2025, NDVI values decreased and became more variable, with larger areas of lower vegetation density.

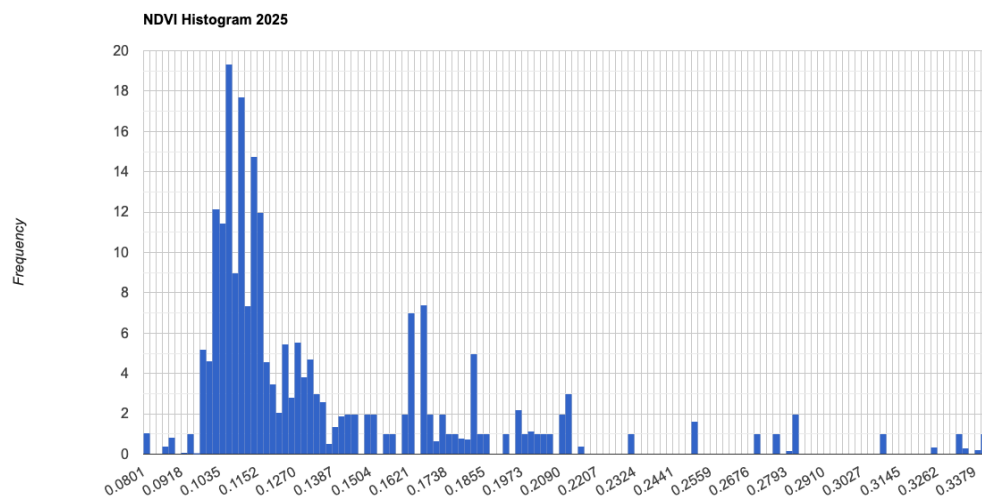
Histograms of NDVI values further support this trend, showing a shift toward lower NDVI values over time, indicating a general decline in vegetation health across the parcel.



[Figure 3] - 2005 NDVI Histogram



[Figure 4] - 2015 NDVI Histogram



[Figure 5] - 2025 NDVI Histogram

Discussion

The results suggest a decline in vegetation condition on the parcel over the past 20 years, which may indicate reduced land productivity and potential challenges for agricultural use. The relatively low NDVI values (0.08–0.3) suggest that vegetation is generally sparse and may be limited by factors such as soil conditions, water availability, or land management practices.

Despite this decline, the presence of moderate NDVI values in some areas indicates that the parcel may still have potential for targeted regeneration. Areas with relatively higher NDVI could represent zones of opportunity for future agricultural or ecological improvement.

Several limitations should be considered. NDVI provides an indirect measure of vegetation health and does not capture soil quality, water rights, irrigation infrastructure, or management history, all of which are critical for determining true agricultural feasibility. Additionally, differences in pixel alignment and data processing between platforms may introduce minor variability in calculated statistics.

Future work could incorporate additional indices such as NDMI (moisture), higher-resolution imagery, or field-based observations to better understand site conditions. Further analysis could also examine seasonal variation or compare the parcel to nearby reference sites.

Conclusion

It is recommended that ECSD begin taking steps to regenerate vegetation on the parcel through improved land management or restoration strategies. In the short term, ECSD should consider partnering with local ranchers or producers to meet food or grazing needs while restoration efforts are underway. This approach supports local agricultural systems while allowing time for long-term improvements to the parcel.